Rule context constrains the interchangeability of content-matched memory states

Summary

Working memory retains information for use under task rules. If memory states have the same content and similar representations, can we use them interchangeably in subsequent computation? Six independently initialized recurrent networks were trained on a permutation-based memory task. Each network received a permutation of 1 through 7, processed operation cues separated by delays, and answered queries about the resulting permutation. The operations included inversion and complementation (replacing each position label j with $8 - j$). A preceding context cue determined whether A meant inversion and B complementation, or vice versa. Exact permutation relations allowed different initial inputs and histories to reach identical current permutations. Within each network, we replaced the memory component of the receiver's hidden state with the donor's while retaining the receiver's control component, subsequent inputs, and readout. Exchanges within the same rule context introduced new errors on 1.7% of all queries, compared with 53.4% across contexts. Under identical subsequent inputs, answers after exchange matched the donor's own answers on $> 99.99\%$ of queries, including nearly all donor errors. Some paired control states were identical. Translating subsequent cues into their donor-context equivalents increased cross-context accuracy from 42.1% to 97.9% on held-out permutations; the same translation without exchange reduced original-task accuracy from 97.9% to 42.1%. Linear centered kernel alignment (CKA) distributions largely overlapped between within-context and cross-context pairs (medians approximately 0.892 and 0.889). Adding CKA to a baseline accounting for task conditions, native performance, content decoding, and input/control-state distances did not consistently improve loss predictions for held-out networks and condition pairs. These results suggest that content-matched memory states can carry different information about how subsequent cues are interpreted. In these networks, matching content and high CKA did not guarantee functional interchangeability, motivating causal tests when interpreting neural population representations.

Additional Detail

The intervention uses two trials in the same trained network (Fig. 1A). Their initial permutations and cue histories can differ while producing exactly the same current permutation. The architecture partitions the hidden state into memory m and control c. We replace m_R with m_D in native coordinates, then use the receiver's subsequent inputs and the unchanged network readout. Retaining c_R specifies a component intervention; it does not establish that the receiver's functional rule is retained. Some paired control states are exactly equal. Intervention definitions and scoring matter for interpretation [2].

New errors count native-correct, exchange-wrong answers divided by *all* evaluated queries (Fig. 1B; 134 B1 permutations, 19 orbits). Means weight networks equally; within each network, condition pairs, both directions, delays $d = 0, 10$, and suffixes NN/NA/NB retain their original weights. Each permutation has seven item-to-position and seven position-to-item queries. Cue translation swaps A/B and preserves N. Both treatments in Fig. 1C are scored against the *original* receiver target, including NN. The no-exchange control is native, unpatched activity. Its 362 B_reserved permutations (51 orbits) were held out from earlier B1 exploration; this submission reuses saved results.

Linear CKA [1] uses content-aligned memory states from the separate geometry set G. Figure 1D uses only $d = 0$, with each network–unordered-condition pair counted once across suffixes. Its pooled median is not the mean of network medians. The original held-out-network/unseen-pair predictor comparison gave mean MAE improvement -0.0149 nat with CKA (one of six folds positive); the original refit-bootstrap 95% interval was very wide, $[-42.56, 87.78]$ nat. This limits precision and does not show that CKA is generally uninformative. Network 106 and weak native conditions remain included. Registered GPU checks passed; the original strict CPU float32 logit tolerance failed (float64 agreed; compared argmax answers were unchanged). These findings concern the tested networks and continuations.

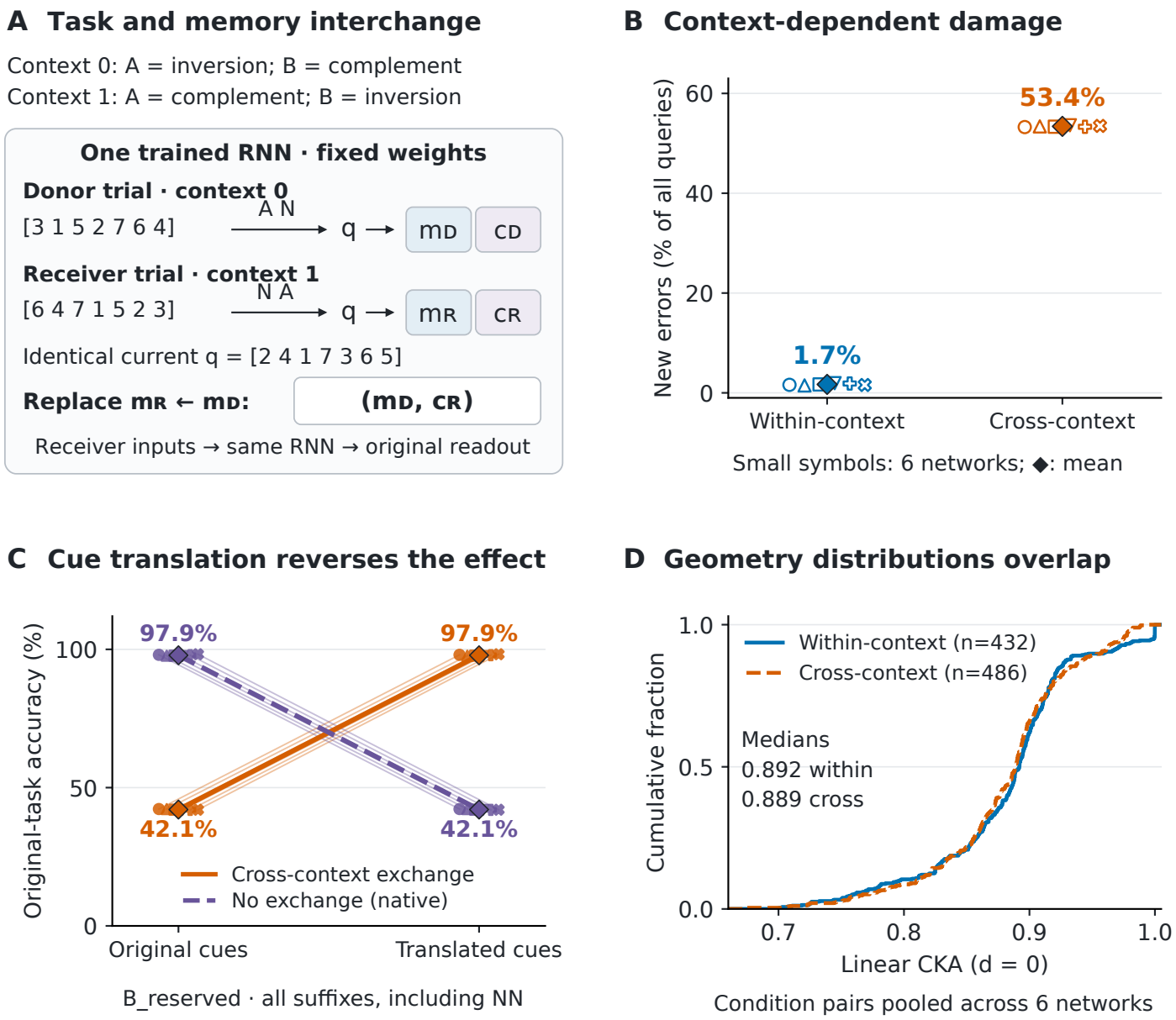


Figure 1. Rule context changes the functional use of matched memory. A, Two trials in one fixed network reach the same full permutation q from different initial inputs and histories (N is a no-operation cue). The exchanged state is (m_D, c_R); keeping c_R does not imply retaining the receiver's functional rule, and c_D need not differ from c_R. **B,** New-error percentages use all B1 queries as denominator; small symbols show six independently trained networks (seeds 101–106), and diamonds show their equal-weight means. **C,** On B_reserved, translating cues compensates cross-context exchange but impairs native, unexchanged performance on the original target. Thin paired lines/small symbols show networks; thick lines/diamonds show equal-weight means. Seed symbols in B/C follow the same order: circle, up-triangle, square, down-triangle, filled plus, X; horizontal offsets separate overlapping marks. All prespecified suffixes, including NN, and both delays contribute to B/C. **D,** ECDFs pool 432 within-context and 486 cross-context geometric measurements at $d = 0$ (72/81 unordered condition pairs per network), with no suffix duplication. These are dependent condition-level distributions across six networks, not 918 independent training runs. Blue/orange denote within/cross context in B/D. Identical future inputs gave $> 99.99\%$ donor-answer agreement, including nearly all donor errors.

References

[1] Kornblith, S., Norouzi, M., Lee, H. & Hinton, G. (2019). Similarity of Neural Network Representations Revisited. *PMLR* 97, 3519–3529. proceedings.mlr.press/v97/kornblith19a.html

[2] Zhang, F. & Nanda, N. (2024). Towards Best Practices of Activation Patching in Language Models: Metrics and Methods. *ICLR*. arXiv:2309.16042.